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Q.No

Q.No	

Q.No

1 1A

Etheruim is a distributed ledger techology.

The types of wallets are
- Non Deterministic wallet
- Deterministic wallet
- Hyerachical Deterministic wallet.

Non Deterministic wallet
- multiple seed phase.

Deterministic wallet
- Single seed phase

Hierachical Deterministic wallet
- Single seed phase and wallet is organize

1B

Hierarchical Deterministic wallet derives multiple address from a single seed

As it can store multiple account It will do it

Q.No

1C It is better to backup the seed phase in a deterministic wallet as it's sufficient to regenerate all private keys

In a deterministic wallet it doesnt need multiple seed phases, It can save a single seed phase which can store your whole etherium account. It is better because it is sufficient to regenerate all private keys

1D EVM

A virtual machine is a an whole ~~operating~~ operating system which is containerized so It can be portable and shared.

An Etherium virtual machine cross checks and executes smart contracts to verify, validate the transaction.

It is like ~~the~~ node of a block chain as if it is verified then only any operation can occur for which It is checked by the whole network

Q.No

1E Keccak 256

Keccak256 is a hashing algorithm used for Ethereum
It is used in smart contracts, account ~~I~~ Ids and
many more
It is also used for the creation of a unique
Id for a user as for it's account.

1F

Wallet	CLI.
• Options for physical wallet.	• No physical options
• Easy to setup	• Hard to setup.
• Accessibility.	• Not that accessible.
• Simple	• complex

Q.No

2

2A

Ethereum network types.

For Ethereum to be active and flowings it needs its network to be active. and alive.

Ther are 3 types.

- Mail Network
- Test network
- Layer 2 Ethereum classic.

-> mail network use case -> communication

The mail network is used for its communation between peers.

This network can be used for communication

Q.No

→ Test network. use case → debugging

The test network is used for debugging and finding any vulnerability or issues before hand. ~~or~~

It can also be used for programming / developing on other types of ethereum or a single branch of it which may have other principles at focus.

→ Ethereum classic ~~→~~ use case → stability.

The Ethereum classic is used for quick setup of a network quickly.

It is a stable version without many issues and if anyone would like to fast setup a network they could use this.

Q.No

2B) Ethereum.

3) Assymetric encryption.

In Asymmetric encryption ~~both~~ the sender and reciever ~~which~~ will have a pair of keys. private key and public key.

If a sender would want to send his message, First he would accquire the reciever's public key and encrypt the message with it.

Then the newly encrypted message will be share all accross till it is recieved by the reciever. Only the reciever can read the message.

Q.No

When the reciever recieves the encrypted message, he will decrypt it with is own private key. This private key is kept hidden and unkown to any One.

Like this a public address is generated by ethereum account ~~from~~ a private key.

This is to ensure If any activite isto be done by the public address It can be traced back to the ethereum account ensuring transparency.

This process ensures only the account owner can Send ether, because so that any secondary account is view only maintaing ~~acc~~ sole access. Only the account owner can send ether as only him only has access to the private and public key

Q.No

2C] Keccack 256

keccack 256 can verify that a given transaction hash matches its data.

For every transaction padding will be added. which contains the trancation data.

We can ~~data~~ cross verify the hashes of it's padding and see what accentally happen.

With in this ~~ev~~ every transaction can be verified and validated.

This helps because so no one can be fooled.

Q.No

2D Remix

- Enable the metamask extension in Remix IDE.

- Create a new Enviornment In Remix IDE

- Develope Code In The Ide and make sure to save the file

- Complie The code to ensure no errors.

- If the code of Smart Contract is executed without errors then we can add github for version control.

- Deploy the Smart Contract

Q.No

4 4A

Ethereum is a distributed ledger technology

There are 3 types of wallets.
- Non Deterministic.
- Deterministic.
- Mismatched Deterministics

→ Non Deterministic Wallets

In the 3 type of etherium wallets, this is on of the types.
In Non Deterministic wallets we would need to Store all the seed phrases and types of It to store your account.

Real world benefit → Storage.

Q.No

Deterministic wallet

In a deterministic wallet: you can just store your seed phrase and It will store your account

Real world benefit → Security

Hierarchical Deterministic wallets

In a Hierarchical Deterministic wallet, It can store multiple accounts and It will be stored in a ~~neat~~ ordered manner.

Real world benefit → multiple accounts.

Q.No

4 B/

Hierarchical deterministic wallet.

In a Hierarchical deterministic wallet It can store multiple accounts and Organized them.

It all needs is from a single master seed.

Using a single master seed It can generate multiple public address.

Even though multiple public address only the private key holder ~~can~~ can mange the account.

Q.No

6 . 6 A Solidity.

Allows a contract owner to set and retrieve an ENS record.

An ENS is Ethereum name system.

code

function getENS(ENS, string 256)

function setENS (ENS, String 256)

This code is for a contract owner to set and retrieve an ENS record.

ENS registry Smart Contract would be queried in the process, to cross verify from ENS to publice address. to hash check it.

Q.No

6B

- Enable the metamask extention in Remix IDE
- Create a new enviornment in remix IDE
- Develope ~~ae~~ your desires code for smart contra
- Make sure to save your file
- Complie your code to ensure no errors
- If code of smart contract is executed without errors, we can implement git for safe version control.
- Deploy the smart contract code.

Q.No

- ~~Upate~~ updat. your code with appropriate chges.
- Make sure to update in git for safe version control and easy roll back.
- Redeploy the smart contract code.

Q.No